

The Fed Put under General Equilibrium

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Literature and Contribution

Question

Should monetary policy react to asset prices?

What changes once this reaction is modeled as a downside insurance?

- The literature has mostly studied symmetric asset-price terms in Taylor rules
- It has documented the empirical Fed-put object as explicitly asymmetric.

This proposal adds

- A closed-form result that rationalizes costs and benefits for this policy
- Numerical exercises for the asymmetric rule

Related work: Bernanke–Gertler; Gilchrist–Leahy; Bullard–Schaling; Carlstrom–Fuerst; Cecchetti et al.; Nisticó; Ifrim; Cieslak–Vissing-Jorgensen; Pastore.

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A NK-Lucas Tree setup

We want to keep "the market" simple

- Start from a textbook NK model with demand and technology shocks
- Add a risky asset with price M_t and dividends D_t
- Dividends are lump-sum income, and do not alter the household's real allocation
- The household's SDF prices the asset

Asset pricing condition

$$m_t = Q_t \mathbb{E}_t [\Pi_{t+1} (m_{t+1} + d_{t+1})].$$

$$d_t = (1 - \rho_d) d^{ss} + \rho_d d_{t-1} + \varepsilon_t^d.$$

→ Real shocks move asset prices through discount rates; dividend shocks are purely financial shocks in the model.

Monetary rule asymmetry and equilibrium

The IS and NKPC are standard, and we add an Asset-Pricing equation to the system:

$$\text{IS : } \tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma} (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) \quad (1)$$

$$\text{NKPC : } \pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t \quad (2)$$

$$\text{AP : } \tilde{m}_t = \mu \mathbb{E}_t [\tilde{m}_{t+1}] + (1 - \mu) \mathbb{E}_t [\tilde{d}_{t+1}] - (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) \quad (3)$$

Monetary policy is subject to shocks, and may be symmetric or put-like. Let $\psi_t = \min\{0, \tilde{m}_t\}$ represent asset-price shortfall. Then,

Symmetric MR

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + \phi_m \tilde{m}_t + u_t$$

Asymmetric MR

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + \phi_m \psi_t + u_t$$

The second rule ignores windfalls ($\tilde{m}_t > 0$)

Solving the symmetrical case

By solving $\tilde{y}_t$ and $\tilde{\pi}_t$ in the form

$$\tilde{y}_t = A_r r_t^n + A_u u_t + A_d d_t \quad (4)$$

$$\pi_t = B_r r_t^n + B_u u_t + B_d d_t \quad (5)$$

We get, with $\Delta(x)$ increasing in ϕ_m

$$A_r = \frac{(1 - \beta\rho_r)(1 - \mu\rho_r)}{\Delta(\rho_r)} \quad B_r = \frac{\kappa(1 - \mu\rho_r)}{\Delta(\rho_r)}, \quad (6)$$

$$A_u = -\frac{(1 - \beta\rho_u)(1 - \mu\rho_u)}{\Delta(\rho_u)} \quad B_u = -\frac{\kappa(1 - \mu\rho_u)}{\Delta(\rho_u)} \quad (7)$$

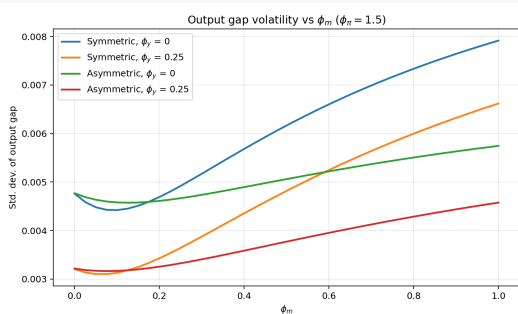
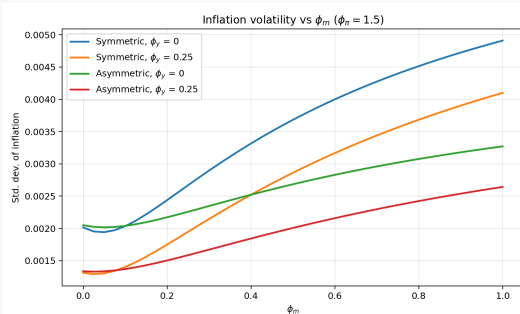
$$A_d = -\frac{(1 - \beta\rho_d)\phi_m(1 - \mu)\rho_d}{\Delta(\rho_d)} \quad B_d = -\frac{\kappa\phi_m(1 - \mu)\rho_d}{\Delta(\rho_d)} \quad (8)$$

$\rightarrow \phi_m$ dampens natural-rate and monetary shocks, but gives dividend shocks nonzero macro effects: the same coefficient stabilizes real shocks and imports financial shocks.

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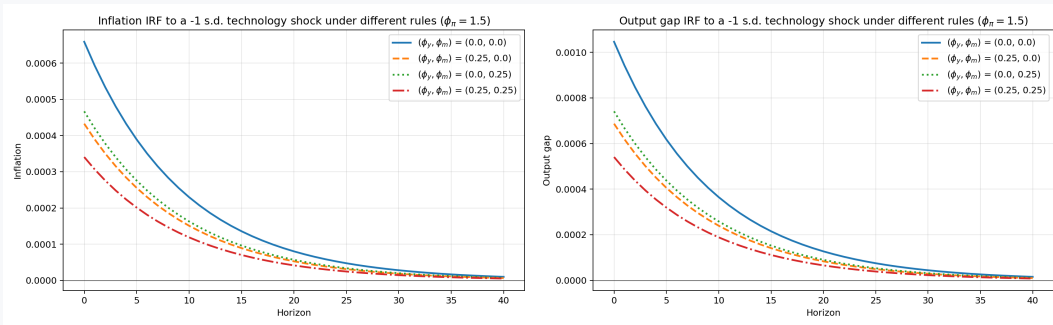
U-shaped volatilities



- A small ϕ_m can initially reduce volatilities
- Financial contamination effect eventually dominates
- The asymmetric rule delivers flatter curves

asset prices

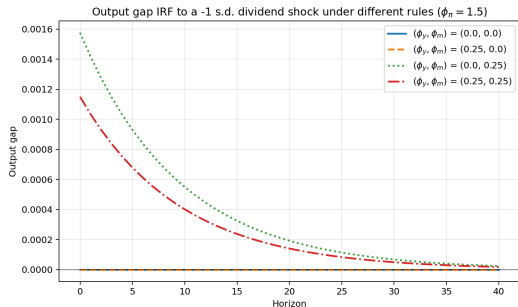
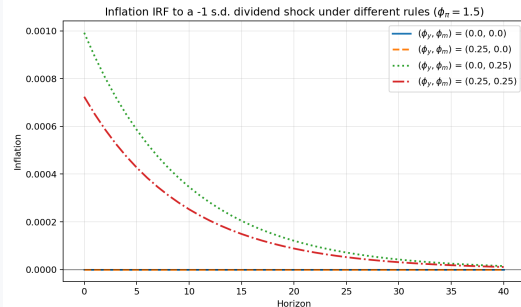
IRFs help us make sense of the result



- Asset prices move with the real discount-rate channel
- Reacting to $\tilde{m}_t$ is in part reacting to the real economy
- Inflation and output gap responses are dampened in the same way as reacting to $\tilde{y}_t$

demand shock

Financial contamination



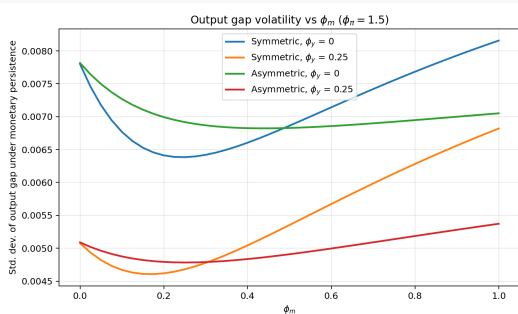
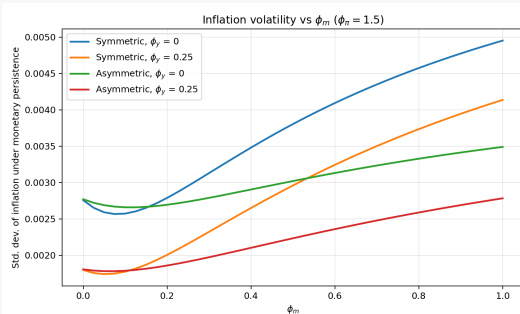
Tradeoff

Dividend shocks are macro-neutral at first. Only once the central bank reacts to asset prices do they generate movements in $\tilde{\pi}_t$ and $\tilde{y}_t$.

Monetary persistence exacerbate the U-shapedness

Until now we have not allowed for monetary persistence. In the real world it is very persistent!

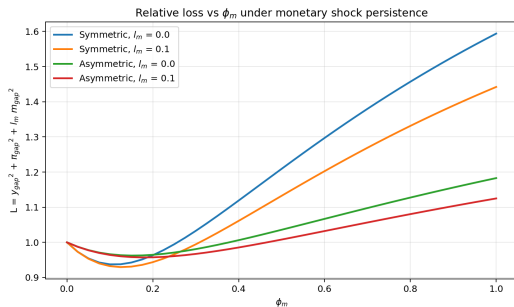
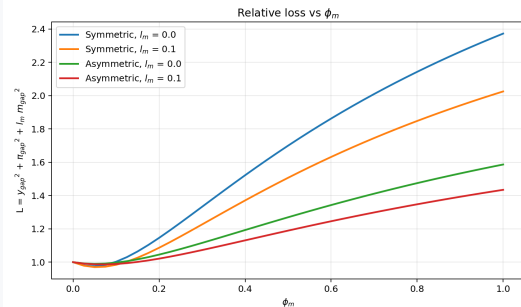
We add it into the model by setting $\rho_u = 0.75$:



→ The qualitative ranking remains, but the payoff from mild asset-price reactivity becomes starker.

Aggregating deviations into loss functions

- Taking both inflation and output into account, what is the best ϕ_m ?
- What if the central bank also wants to deliver asset price smoothing?



→ Even when $l_m = 0$, small ϕ_m are at worse neutral

→ For a very low l_m , small ϕ_m delivers smaller losses

loss under $\phi_y = 0$

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Taking stock

The conventional reading:

Direct asset-price reactions are undesirable: they worsen determinacy and/or add irrelevant financial noise to monetary policy

We argue instead:

The conclusion depends on the information environment and on whether the rule is symmetric. Small reactions can be useful when asset prices reveal real shocks, especially when output gap is unavailable or poorly measured.

Core interpretation

High ϕ_m looks like a risk transfer from financial assets to the real economy. Low ϕ_m , especially under the asymmetric rule, can instead be read as limited information use plus downside stabilization.

Further work

Finish the baseline model

- Derive the precise equivalence between ϕ_m and ϕ_y for real shocks.
- Characterize optimal $(\phi_\pi, \phi_y, \phi_m)$ under a stated loss function.
- Study determinacy under asset-price terms, in the spirit of Bullard–Mitra/Bullard–Schaling.

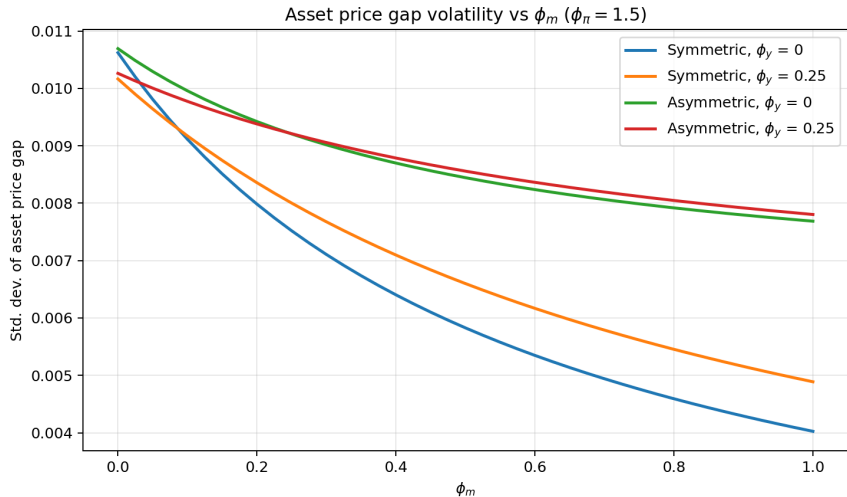
Extend the mechanism

- Replace the Lucas tree with equity claims on firm profits.
- Add persistence or stochastic volatility in asset-price shocks.
- Introduce agent heterogeneity or OLG structure to create a direct wealth-effect channel.

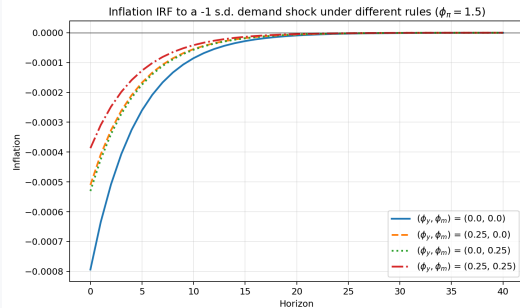
Open question

Can an empirically plausible wealth channel turn the Fed put from an information device into a genuine aggregate-demand stabilizer?

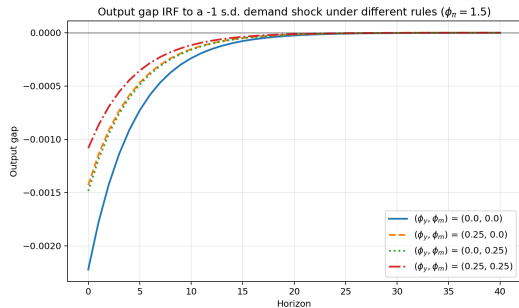
Appendix: Asset price volatility



Appendix: Demand shock IRFs



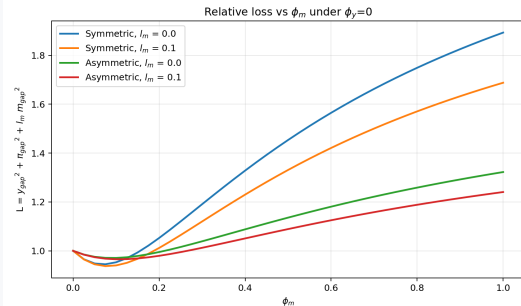
Inflation



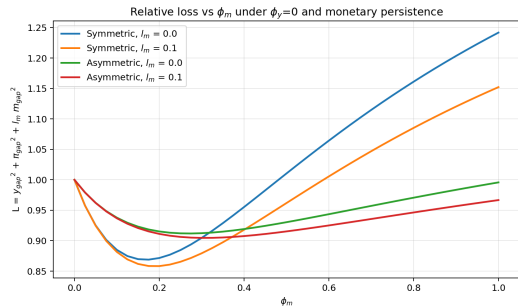
Output gap

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Appendix: Loss absent ϕ_y



Inflation



Output gap

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